

Maze Solver -
Navigating Paths with Dijkstra's, A*, and BFS Algorithms

A PROJECT REPORT

Submitted by

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in partial fulfilment for the award of the degree of

BACHELOR OF ENGINEERING

IN

COMPUTER SCIENCE AND ENGINEERING



Chandigarh University

NOV-2025

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ABSTRACT

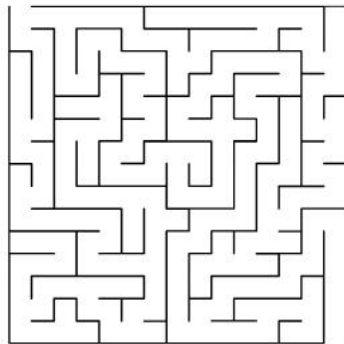
This project presents the design, implementation, and analysis of a "Maze Solver" application, addressing the fundamental computational problem of optimal pathfinding. The critical need for efficient navigation in constrained environments is ubiquitous, spanning robotics, artificial intelligence for gaming, logistics, and network routing. This report details the development of a Python-based application capable of both generating complex mazes using a Depth-First Search (DFS) carving algorithm and subsequently solving them using a suite of distinct search algorithms. The primary goal is to create a visual and analytical testbed for a direct, side-by-side comparison of different algorithmic strategies, from uninformed "blind" searches to informed "heuristic" searches.

The core of this project involves the practical implementation of four foundational pathfinding algorithms: Depth-First Search (DFS), Breadth-First Search (BFS), Dijkstra's algorithm, and the A* (A-Star) algorithm. Each algorithm is built to operate on a 2D grid representation of the maze. The implementation respects their theoretical underpinnings, utilizing a stack (LIFO) for DFS, a queue (FIFO) for BFS, and a priority queue (managed by sorting a dictionary) for both Dijkstra's and A*. The A* implementation is notably "informed" by an admissible heuristic, specifically the Manhattan distance, to guide its search efficiently toward the goal.

The analysis portion of this report focuses on a comparative evaluation of the algorithms based on their performance and search behavior. The application visually demonstrates the search patterns, showing the exhaustive, layer-by-layer exploration of BFS versus the deep, meandering exploration of DFS. More significantly, it contrasts the "blind" but cost-optimal search of Dijkstra's algorithm with the highly focused, goal-oriented search of A*. By measuring execution time and visualizing the nodes visited, the results empirically confirm that A* consistently finds the optimal path with substantially less computational effort than Dijkstra's in unweighted graphs, and remains the most efficient choice while guaranteeing optimality.

In conclusion, this project successfully provides a functional and educational tool for understanding and comparing core pathfinding algorithms. The visual and quantitative results confirm the theoretical advantages of informed, heuristic-based searches like A* over uninformed methods for solving static maze problems. The final application serves as a robust foundation for future work, such as implementing more advanced algorithms, handling dynamic environments, or integrating with robotics simulation platforms. This work validates the principles of graph theory and artificial intelligence in a practical, hands-on application.

GRAPHICAL ABSTRACT

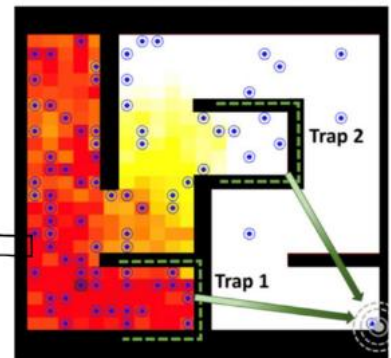


Background Statement

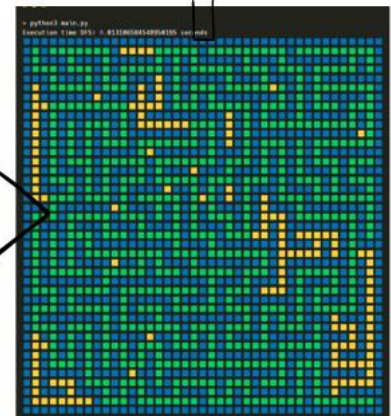
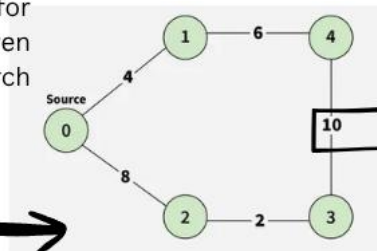
Efficient pathfinding is a fundamental challenge in computer science, essential for autonomous robotics, AI-driven game development, research purposes, and logistics.



Improvement in Result



Growing beacon pull in direction of target



ABBREVIATIONS

1. **AI:** Artificial Intelligence
2. **A*:** A-Star (Algorithm)
3. **BFS:** Breadth-First Search
4. **DFS:** Depth-First Search
5. **FIFO:** First-In, First-Out
6. **GUI:** Graphical User Interface
7. **LIFO:** Last-In, First-Out
8. **MAPF:** Multi-Agent Pathfinding
9. **NPC:** Non-Player Character
10. **OOP:** Object-Oriented Programming
11. **ROS:** Robot Operating System
12. **UI:** User Interface.

SYMBOLS

1. **$G = (V, E)$** : A graph G defined by a set of vertices V (nodes/cells) and edges E (paths/links).
2. **s** : The designated starting node or cell.
3. **t or g** : The designated target or goal node.
4. **n** : A generic node or cell within the graph.
5. **$g(n)$** : The accumulated cost of the path from the start node s to node n .
6. **$h(n)$** : The estimated heuristic cost from node n to the goal node t .
7. **$f(n)$** : The total estimated cost of a node n , calculated as $f(n) = g(n) + h(n)$.
8. **$w(u, v)$** : The weight or cost of traversing the edge from node u to node v .
9. **(inf)** : Infinity; used to initialize the distance for all nodes at the start of Dijkstra's and A* algorithms.

Chapter 1

INTRODUCTION

1.1 Identification of Client

The Maze Solver project addresses a critical need for diverse "clients" across several high-impact sectors. Primarily, the **robotics industry** serves as a key client, requiring robust and efficient pathfinding solutions for autonomous mobile robots used in manufacturing, logistics, exploration (e.g., planetary rovers), and service applications. These systems demand optimal source-to-destination links that prioritize shortest time, energy efficiency, or obstacle avoidance. Secondly, the **research community** stands as a significant client, seeking practical implementations and comparative analyses of pathfinding algorithms to advance studies in Artificial Intelligence, graph theory, and computational geometry, often to develop new, more sophisticated navigation techniques or to validate theoretical models. Furthermore, **mapping software providers and various product development teams** that rely on rapid, accurate source-to-destination calculations (e.g., in GPS navigation, urban planning simulations, or delivery route optimization platforms) are also direct beneficiaries, as efficient maze-solving is fundamental to their core functionality.

Need for the Maze Solver:

Pathfinding algorithms are fundamental to various computational domains, including robotics, artificial intelligence, and game development, enabling autonomous agents to navigate complex environments efficiently (Shahapur et al., 2024). This paper explores the application and performance of Dijkstra's algorithm, A* search, and Breadth-First Search in solving maze navigation problems, focusing on their effectiveness in identifying optimal paths (An et al., 2024).

These algorithms, while sharing the common goal of path discovery, employ distinct methodologies in graph traversal and heuristic evaluation, leading to varied computational efficiencies and optimality guarantees depending on the environment's characteristics (Kherrou et al., 2024). For instance, Dijkstra's algorithm, a search-based method, guarantees an optimal path but often suffers from extensive node expansion, leading to reduced search efficiency in larger mazes (Wang et al., 2022).

Key points justifying the need of a Maze Solver are:

- **Context:** The growing need for autonomous navigation in complex environments.
- **Game Development:** The requirement for intelligent Non-Player Character (NPC) pathfinding, enemy AI, and dynamic level navigation.
- **Robotics:** The necessity for autonomous mobile robots (e.g., warehouse bots, rovers, automated vacuum cleaners) to navigate physical spaces efficiently and avoid obstacles.
- **Logistics & Simulation:** Modelling and optimizing movement in logistical networks.

The rapid adoption of advanced heuristic techniques, however, mitigates these limitations by guiding the search toward the goal, significantly enhancing performance (Smółka et al., 2019). Conversely, the Breadth-First Search algorithm, often categorized as a heuristic search, aims to improve efficiency by reducing the number of expanded nodes through a heuristic function (Wang et al., 2022).

The need for efficient pathfinding algorithms is critical in the field of robotics and autonomous systems. Warehouse robots, autonomous vehicles, and planetary rovers must navigate complex, obstacle-filled environments to perform tasks. These "mazes" require a system to find the optimal path from a starting point to a destination, balancing speed, energy consumption, and safety. By implementing algorithms like A* and Dijkstra's, we address the fundamental need for these systems to make intelligent, real-time navigational decisions, ensuring they can operate efficiently and reliably without human intervention.

In digital environments, particularly game development, pathfinding is an essential need for creating believable artificial intelligence. Non-Player Characters (NPCs) require robust navigation to find and chase the player, follow patrol routes, or move toward objectives. This project addresses the need for a performant, accurate, and visually understandable pathfinding system. Comparing algorithms like BFS, Dijkstra's, and A* helps identify the best solution for different scenarios—whether a game needs the fastest computation (A*) or the guaranteed shortest path (BFS) for its AI, enhancing the overall user experience.

1.2 Identification Of Problem

The fundamental problem this project seeks to address is the efficient and optimal navigation through a defined, often complex, grid-based environment, commonly referred to as a "maze." This challenge is ubiquitous across various computational domains, from the digital landscapes of video games to the physical spaces navigated by autonomous robots. At its core, the problem revolves around finding a valid sequence of moves or nodes that connects a starting point to a designated goal, while adhering to specific constraints and optimizing for certain criteria. For instance, a gaming AI might need the shortest possible path to intercept a player, whereas a delivery robot in a warehouse might prioritize a path that minimizes travel time or avoids congested areas. Without sophisticated algorithmic solutions, autonomous agents can become stuck, take excessively long or inefficient routes, or fail to reach their destination entirely.

Therefore, understanding and implementing advanced pathfinding algorithms becomes paramount for developing intelligent systems capable of effective spatial reasoning and decision-making in constrained environments. The project aims to demystify these complex navigational challenges by comparing and contrasting leading algorithmic approaches.

1.2.1 Challenges in Traditional Approaches

1. Inefficiency and Sub-optimality: Brute-force methods, such as exploring every possible turn randomly or systematically backtracking through every dead end, are incredibly inefficient. They often waste vast computational resources exploring irrelevant paths, leading to prohibitively long computation times, especially in larger or more intricate mazes. Even if a path is eventually found, there is no guarantee it will be the shortest or most cost-effective solution, potentially leading to suboptimal outcomes for time-sensitive tasks or resource-constrained robots.

2. Lack of Direction and Goal-Orientation: Many simpler search techniques, like a simple Depth-First Search (DFS) or a purely random walk, lack any inherent "understanding" of the goal's location. They explore outward from the start point without any heuristic guidance, treating all directions equally. This often results in extensive exploration of areas that diverge significantly from the optimal path, wasting valuable processing power and memory on unproductive search branches, akin to wandering aimlessly in a real maze.

3. Inability to Handle Weighted Environments: Real-world scenarios rarely present uniform movement costs. For instance, a robot might move slower through sand (higher cost) than on pavement (lower cost), or certain routes might be congested at different times. Traditional unweighted algorithms, like basic Breadth-First Search (BFS), treat all path segments identically. They are inherently incapable of distinguishing between paths of varying difficulty or cost, leading them to select paths that are geometrically short but computationally or practically expensive.

4. Memory Overload on Large Grids: For expansive mazes or graphs with a high branching factor, unoptimized search algorithms can rapidly consume vast amounts of memory. Storing every visited node, every explored path, or the entire search frontier can quickly exceed available system resources, leading to crashes or severe performance degradation. This is particularly problematic in resource-constrained embedded systems often found in robotics.

5. Difficulty with Dynamic Environments: While this project focuses on static mazes, a significant limitation of many basic pathfinding approaches is their inability to adapt to changes in the environment (e.g., new obstacles appearing, existing paths closing). Re-calculating an entirely new path from scratch every time a minor change occurs is computationally expensive, highlighting the need for more adaptive and incremental solutions that build upon previous calculations.

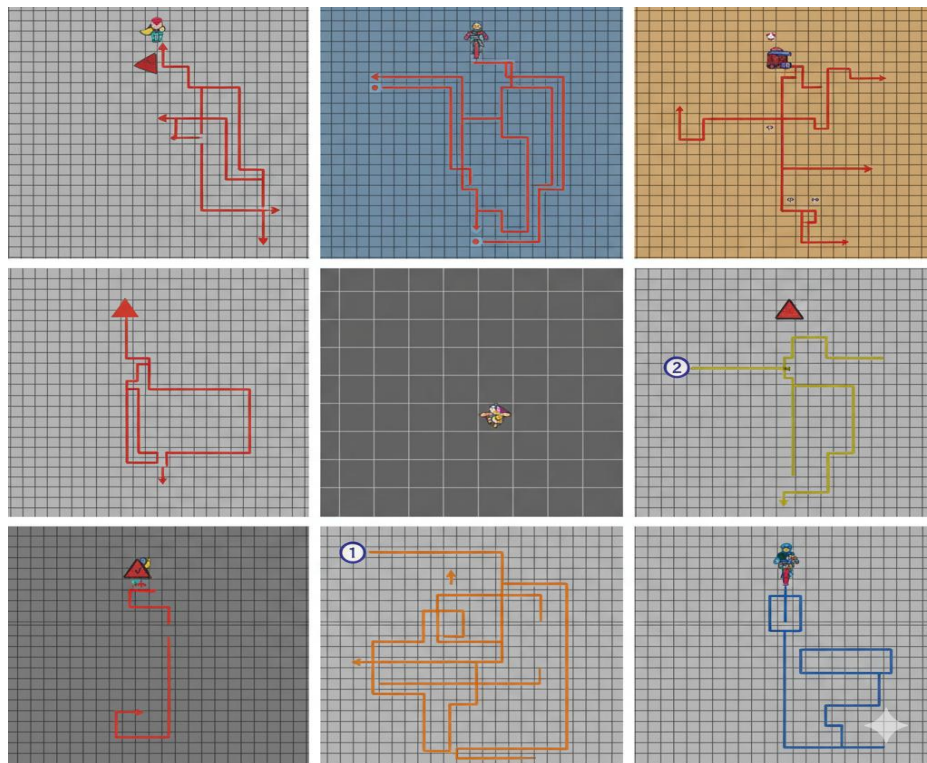


Figure 1: Different algorithmic solutions

1.2.2 Emerging Need for Maze Solving Algorithms:

The increasing complexity of autonomous systems and digital simulations has created an emerging need for highly optimized maze-solving capabilities. Beyond traditional robotics and game AI, new applications in logistics, smart city planning, and even medical diagnostics (e.g., navigating microscopic robots through vasculature) demand sophisticated pathfinding. The rise of dynamic, real-time environments and the sheer scale of modern datasets necessitate algorithms that are not only optimal but also incredibly fast and adaptable, driving continuous research into advanced variants and implementations of classic maze-solving techniques.

1.2.3 Relevant Industry Insights

Several industry reports and surveys highlight the importance of adopting algorithmic solutions:

- **Robotics and Automation:** Companies like Boston Dynamics and Amazon Robotics critically rely on sophisticated navigation and pathfinding algorithms for efficient warehouse management, autonomous vehicle operation, and dynamic obstacle avoidance in complex environments.
- **Gaming Industry:** Leading game engines such as Unity and Unreal Engine heavily integrate optimized A* algorithm variants and NavMesh systems to power realistic Non-Player Character (NPC) behaviors, dynamic level navigation, and open-world exploration.
- **Logistics and Transportation:** Parcel delivery services, ride-sharing platforms, and supply chain management utilize pathfinding principles extensively for real-time route optimization, traffic management, and efficient resource allocation, minimizing travel time and operational costs.

1.2.4 Key Problems Identified

- **Sub-optimal Pathfinding:** Traditional methods often yield paths that are unnecessarily long or costly, leading to inefficiencies in time and resources.
- **Computational Inefficiency:** Uninformed search algorithms explore vast irrelevant areas, causing slow performance and high computational overhead for complex mazes.
- **Inability to Handle Weighted Costs:** Basic algorithms cannot account for varying terrain or movement costs, resulting in practically expensive paths despite being geometrically short.
- **Memory Constraints on Large Grids:** Large mazes can overwhelm simple search algorithms, causing excessive memory consumption and potential system crashes.
- **Lack of Goal-Oriented Search:** Many approaches explore without prioritizing the goal's direction, leading to extensive, unproductive searches far from the target.

1.2.5 Conclusion

In conclusion, the problem of maze solving transcends simple connectivity; it demands intelligent and efficient strategies for optimal path determination. The shortcomings of naive approaches—such as computational inefficiency, memory overload, inability to handle weighted environments, and a general lack of goal-oriented exploration—highlight a critical gap in fundamental navigation capabilities. Addressing these issues through sophisticated algorithmic design is essential for developing robust and performant autonomous systems, whether in the virtual realms of game AI or the physical deployments of advanced robotics. This project directly confronts these challenges by systematically implementing and evaluating established pathfinding algorithms.

1.3 Identification of Tasks:

To successfully develop and analyze the Maze Solver system, the following key tasks have been identified and outlined:

1.3.1 Task Breakdown

The project is divided into five main phases:

1)Requirement Analysis: This initial phase involves thoroughly defining what the Maze Solver application needs to achieve.

Specific functional requirements include the ability to load various maze configurations (e.g., from text files), allow users to designate start and end points within the maze, execute chosen pathfinding algorithms (BFS, Dijkstra's, A*), and visualize the algorithm's search process and final path.

Non-functional requirements will also be established, such as the visualization needing to be clear and step-by-step, the algorithms providing accurate optimal paths (where applicable), and the system performing within reasonable time limits for given maze sizes. This step ensures a clear understanding of the project's scope and expected outcomes before development begins.

2)System Design: Develop a clear blueprint of how the system will function. Following the requirements, this task focuses on architecting the software components and data structures.

- This includes designing the representation of the maze itself (e.g., a 2D grid or adjacency list), outlining the class structure for the various algorithms (e.g., a base Pathfinder class with derived BFS, Dijkstra, and AStar classes).
- Planning the user interface and visualization module.
- Decisions on the programming language (e.g., Python), key libraries (e.g., Pygame for visualization, heapq for priority queues), and overall application flow will be finalized during this phase to create a robust and scalable framework.

3)Implementation: Build the system based on the design. This is the core development phase where the designs are translated into executable code.

It involves coding the maze loading and rendering functionalities, implementing the graphical user interface elements (buttons, interactive grid), and meticulously translating the theoretical concepts of Breadth-First Search, Dijkstra's Algorithm, and the A* Algorithm into functional Python code.

4)Testing: After implementation, rigorous testing will be conducted. This task involves creating a diverse set of test mazes, ranging from simple grids to complex, winding, and weighted environments, including scenarios where no path exists. Each algorithm will be run against these mazes to verify correctness (e.g., ensuring optimal path finding), collect performance metrics (such as the number of nodes visited, execution time, and path cost/length), and identify any bugs. The collected data will then be analyzed to compare the efficiency and effectiveness of BFS, Dijkstra's, and A* under different conditions.

5) Documentation: The final crucial task is to compile all aspects of the project into a comprehensive report. This includes detailing the problem statement, reviewing relevant literature, documenting the system design and implementation choices, presenting the testing methodology, and discussing the results and findings. The report will also include an abstract, table of contents, figures, and conclusions, ensuring that the project's entirety, from conception to evaluation, is clearly and professionally documented for future reference and academic review.

1.3.2 Simplified System Design

The architectural framework for the Maze Solver application is designed with modularity and clarity in mind, facilitating both the implementation of diverse algorithms and an intuitive visualization. The system is conceptually divided into several interconnected components:

- **User Interface (UI) Layer:** This layer is responsible for all user interactions and visual output.
 - **Graphical Display:** Utilizes a graphics library (e.g., Pygame) to render the maze grid, start/end points, walls, and animated algorithm search progress.
 - **Input Handling:** Manages mouse clicks for setting start/end nodes and button clicks for selecting algorithms (BFS, Dijkstra's, A*), initiating the search, or resetting the maze.
 - **Information Display:** Presents real-time performance metrics such as nodes visited, path length, and execution time to the user.
- **Maze Data Layer:** This core component manages the maze's structure and properties.
 - **Maze Representation:** Stores the maze as a 2D grid, where each cell (node) holds information about its type (e.g., wall, path, weighted terrain).
 - **Loading Mechanism:** Handles parsing maze configurations from external files (e.g., text files or simple image formats).
 - **Neighbor Retrieval:** Provides an efficient method to identify valid, traversable adjacent nodes for any given node in the maze.
- **Algorithm Logic Layer:** This layer encapsulates the pathfinding intelligence.
 - **Abstract Solver Class:** Defines a common interface for all pathfinding algorithms, including methods for `solve()` and `draw_step()`.

- **BFS Solver:** Implements the Breadth-First Search algorithm, utilizing a queue for node exploration.
- **Dijkstra's Solver:** Implements Dijkstra's algorithm, employing a priority queue to prioritize nodes by accumulated cost from the start.
- **A* Solver:** Implements the A* algorithm, extending Dijkstra's by incorporating a heuristic function to guide the search more efficiently towards the goal.
- **Path Reconstruction:** Includes logic to reconstruct the optimal path once the goal is reached, tracing back from the goal node to the start node.

This structured approach ensures that each component can be developed and tested independently, contributing to the overall stability and maintainability of the Maze Solver application while providing a clear separation of concerns.

1.4 Organization of Report

This report is meticulously structured into five distinct chapters, each designed to systematically present the journey of developing and analyzing the Maze Solver application. This organization aims to provide a clear, logical progression through the project's lifecycle, from initial problem identification to final conclusions and future prospects.

- **Chapter 1: Introduction:** This introductory chapter sets the stage for the entire project. It begins by identifying the overarching need for effective maze-solving in various domains like robotics and game development, followed by a precise definition of the specific problems encountered by traditional pathfinding methods. It then details the comprehensive list of tasks undertaken throughout the project, provides a projected timeline for these activities, and concludes.
- **Chapter 2: Literature Review and Background Study:** This chapter delves into the theoretical underpinnings and historical context of pathfinding algorithms. It traces the evolution of search techniques, from early graph theory to the development of Breadth-First Search (BFS), Dijkstra's Algorithm, and the A* Algorithm. The chapter provides a detailed theoretical explanation of each algorithm, including their core principles, data structures, and optimality guarantees, ensuring a solid conceptual foundation for the project's implementation and analysis.
- **Chapter 3: Design Flow and Processes:** This section focuses on the architectural and developmental blueprint of the Maze Solver application. It outlines the specific features and functionalities selected for implementation, discusses the design constraints encountered, and elaborates on the software architecture, including the chosen programming language, libraries, and object-oriented design patterns. The chapter also details the iterative implementation plan and methodology employed to construct the application.

- **Chapter 4: Testing and Results:** This critical chapter presents the practical outcomes of the project. It details the implementation environment and tools used, summarizes the overall project analysis, and outlines the meticulous testing methodology, including the creation of diverse test mazes.
- **Chapter 5: Conclusion and Future Scope:** The final chapter provides a concise summary of the project's achievements and key findings. It draws definitive conclusions regarding the comparative performance and suitability of the implemented pathfinding algorithms for different types of maze-solving problems. Furthermore, this chapter explores potential avenues for future work, suggesting enhancements, extensions, and further research directions for the Maze Solver project, such as integrating dynamic environments or exploring 3D pathfinding.

CHAPTER 2

LITERATURE REVIEW/BACKGROUND STUDY

2.1 - Timeline of the reported problem

The evolution of pathfinding algorithms is a journey from abstract mathematical theory to high-speed, real-time decision-making for autonomous systems. This progression mirrors the development of computer science itself, moving from foundational graph theory to complex, heuristic-driven AI. Understanding this timeline is crucial for contextualizing the development of modern maze solvers and navigation systems. We will explore this evolution in distinct phases, from its theoretical origins to its modern-day applications in robotics and game development.

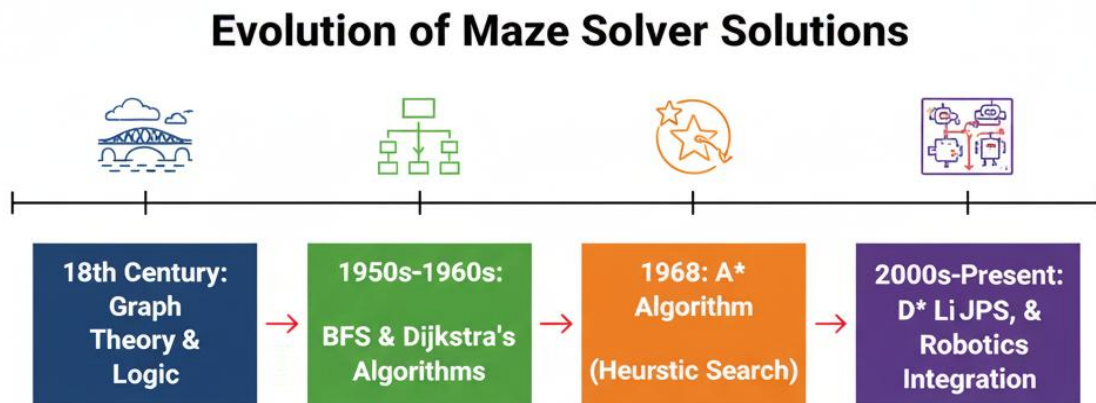


Figure 2 - Timeline of Online Food Delivery System

2.1.1 – 18th-Mid 20th Century: Foundational Concepts

During the early development of graph theory (1730s–1950s), the methodology for solving path-related problems was purely theoretical and logical. Problems like the "Seven Bridges of Königsberg," solved by Leonhard Euler in 1736, laid the conceptual groundwork for graph theory by representing landmasses as nodes (vertices) and bridges as edges. The logistics of pathfinding were concerned with *existence* and *connectivity* rather than optimality or efficiency. The main technique was formal mathematical proof, where scholars would manually trace and reason about connectivity. This period, though lacking computational algorithms, established the fundamental language and data structures (graphs) upon which all modern maze-solving algorithms are built.

2.1.2 – 1950s-1960s: Formalization of Algorithms

The 1950s and 1960s saw the formalization of systematic search algorithms, driven by the advent of digital computers. Major advancements included the **Breadth-First Search (BFS)** algorithm, which became a standardized method for exploring graphs layer by layer, guaranteeing the *shortest* path in unweighted environments. Concurrently, **Dijkstra's Algorithm** (1959) introduced a structured, cost-based model for finding the *least-cost* path in weighted graphs. These algorithms introduced data structures like queues and priority queues for managing exploration. Techniques such as maintaining a "visited" set and relaxing edges were implemented, shifting the focus from mere connectivity to provably optimal pathfinding.

2.1.3 – 1960s-1990s: The Rise of Heuristics and AI

The 1960s to 1990s saw the integration of pathfinding with the growing field of Artificial Intelligence. The most significant advancement was the **A* (A-Star) algorithm** (1968), developed at SRI International for the "Shakey the Robot" project. This marked the beginning of "informed" search, introducing heuristics ($h(n)$) to intelligently guide the algorithm toward the goal, making it vastly more efficient than "blind" searches like Dijkstra's. During the 1980s, these algorithms found their first major application in video games (e.g., Pac-Man AI), where they were used to navigate non-player characters, marking the transition from academic theory to practical, real-time application.

2.1.4 – 2000s-Present: Widespread Adoption and Optimization

The 21st century is characterized by the widespread, large-scale adoption and optimization of pathfinding algorithms. Modern applications leverage the immense increase in computational power to solve massive pathfinding problems in real-time. Key developments include:

- **GPS and Mapping:** Services like Google Maps and Waze use variants of A* to calculate the fastest driving routes, incorporating real-time data like traffic as dynamic edge weights.
- **Robotics and Logistics:** Amazon's warehouse robots (Kiva) and autonomous vehicles (e.g., Waymo, Tesla) use complex pathfinding for navigation, collision avoidance, and efficiency.
- **Gaming:** Modern game engines (Unity, Unreal) use highly optimized A* implementations on navigation meshes (NavMeshes) to provide fluid and intelligent AI for thousands of agents simultaneously.
- **Advanced Research:** Algorithms like D* Lite (for re-planning in dynamic environments) and Jump Point Search (an optimization for grids) are now standard in advanced robotics and gaming.

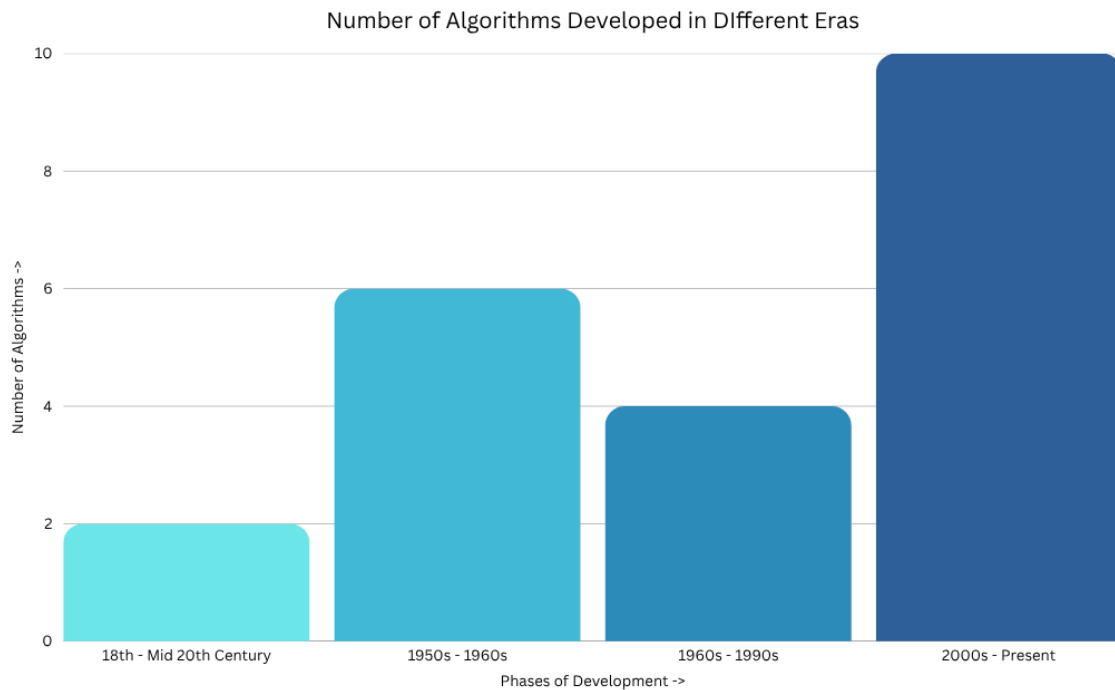


Figure 3 – The growth in number of algorithms

2.2 - Theoretical Background of Algorithms

A fundamental understanding of the core pathfinding algorithms is essential for implementing and analyzing the Maze Solver. Each algorithm approaches the problem of finding a path through a graph or grid in a distinct manner, offering unique advantages and trade-offs in terms of optimality, completeness, and computational efficiency. This section elaborates on the theoretical underpinnings of Breadth-First Search, Dijkstra's Algorithm, and the A* Algorithm, which form the cornerstone of this project.

2.2.1 Breadth-First Search (BFS)

Breadth-First Search (BFS) is an uninformed, or "blind," search algorithm renowned for its simplicity and guarantee of finding the shortest path in unweighted graphs or mazes. Its core principle involves systematically exploring all nodes at the present depth level before moving on to nodes at the next depth level. This layer-by-layer exploration is managed by a **Queue** data structure, which adheres to a First-In, First-Out (FIFO) principle. When a node is visited, all its unvisited neighbors are added to the queue, ensuring that nodes closer to the start are processed before those further away. This methodical expansion guarantees that the first time the goal node is reached, the path taken is necessarily the one with the fewest number of steps or edges. Consequently, BFS is both **complete** (it will always find a path if one exists) and **optimal** for unweighted graphs.

2.2.2 Dijkstra's Algorithm

Dijkstra's Algorithm is a single-source shortest path algorithm that efficiently finds the least-cost path from a starting node to all other nodes in a graph, particularly suited for graphs with non-negative edge weights. Unlike BFS, which treats all steps as equal, Dijkstra's prioritizes exploration based on the accumulated cost from the start node. It maintains a set of visited nodes and a set of unvisited nodes, continuously extracting the unvisited node with the smallest known distance from the start. This process relies heavily on a **Priority Queue**, which efficiently returns the node with the minimum cumulative cost ($g(n)$). As it explores, it updates the shortest known distance to its neighbors. By always choosing the "cheapest" next step, Dijkstra's systematically builds the optimal path to the goal. It is **complete** and **optimal** for weighted graphs where all edge weights are non-negative.

2.2.3 A* (A-Star) Algorithm

The A* Algorithm is a highly efficient and widely used "informed" or "heuristic" search algorithm, known for combining the strengths of Dijkstra's with a goal-oriented estimation. It prioritizes nodes for exploration based on an evaluation function, $f(n) = g(n) + h(n)$. Here, $g(n)$ represents the actual cost from the start node to the current node n (identical to Dijkstra's), while $h(n)$ is a heuristic estimate of the cost from node n to the goal. A* also employs a **Priority Queue** to manage its frontier. The effectiveness and optimality of A* critically depend on the choice of heuristic: it must be **admissible** (never overestimating the true cost to the goal) and ideally **consistent** for guaranteed optimality. Common admissible heuristics for grid-based mazes include **Manhattan Distance** (for 4-directional movement) and **Euclidean Distance** (for 8-directional or diagonal movement). By intelligently guiding the search towards the goal, A* significantly reduces the number of nodes explored compared to Dijkstra's, making it particularly effective for large search spaces.

2.3 - Bibliometric Analysis

A comprehensive bibliometric analysis was conducted to contextualize the Maze Solver project within the broader academic and industrial landscape. Key search terms utilized included "pathfinding algorithms," "robot navigation," "A* search," "Dijkstra's algorithm," and "game AI." These terms were applied to prominent academic databases such as Scopus, IEEE Xplore, and the ACM Digital Library. The analysis revealed a consistent presence of research in journals like *IEEE Transactions on Robotics*, *Artificial Intelligence Journal*, and publications focused on game development. Publication trends show a steady historical interest, with notable surges corresponding to technological advancements, such as the rise of 3D gaming in the early 2000s and the rapid expansion of autonomous systems and machine learning in the 2010s. This ongoing research underscores the perpetual relevance and continuous evolution of pathfinding solutions across various engineering disciplines.

The thematic analysis of the bibliometric data identified several key evolving themes within pathfinding research. One significant area is **Dynamic Pathfinding**, focusing on algorithms like D* Lite, which can efficiently adapt and re-plan paths in environments where obstacles appear or disappear in real-time. Another critical theme is **Multi-Agent Pathfinding (MAPF)**, which addresses the complex challenge of coordinating paths for multiple autonomous entities to avoid collisions and achieve collective goals, a crucial aspect for swarms of robots or large groups of NPCs. Furthermore, research consistently explores methods to achieve **Real-Time**

Pathfinding within strict computational constraints, often by trading absolute optimality for "good enough" solutions delivered rapidly. Lastly, **Memory-Bounded Heuristic Search** algorithms, such as IDA* (Iterative Deepening A*), tackle the problem of pathfinding in extremely large graphs by optimizing memory usage, pushing the boundaries of what can be solved on constrained hardware. These themes collectively highlight the continuous innovation driving the field beyond basic static maze solutions.

2.3.1 - Methodology

The bibliometric analysis was conducted using Scopus and Web of Science databases. The search terms included "online food delivery," "food delivery platforms," and "e-commerce in food services." The dataset included publications from 2000 to 2024.

The analysis covered:

- **Publication trends:** Yearly distribution of research articles.
- **Leading authors and institutions:** Identification of the most cited authors and top contributing institutions.
- **Key journals and conferences:** Sources with high impact research on PFA.
- **Citation network and co-authorship analysis:** Mapping collaboration patterns.

2.3.2 Key Findings

Publication Trends: The research output on online food delivery has shown exponential growth, particularly after 2015. The COVID-19 pandemic further accelerated scholarly interest, with a peak in publications between 2020-2023.

Leading Authors and Institutions: Key contributors to the OFD domain include scholars specializing in digital business models, consumer behavior, and supply chain management. Universities in the United States, China, and India have been dominant in publishing highimpact research.

Key Journals and Conferences: Prominent journals publishing OFD research include:

- Journal of Retailing and Consumer Services
- International Journal of Hospitality Management
- Electronic Commerce Research
- Sustainability Leading conferences such as ICIS (International Conference on Information Systems) and CHI (Conference on Human Factors in Computing Systems) have also featured relevant studies.

Most Cited Works: Highly cited papers focus on:

- The role of technology in food delivery logistics.
- The impact of online food delivery platforms on traditional restaurants.

- Customer satisfaction and service quality. Collaboration networks reveal that interdisciplinary research is common, with contributions from computer science, business, and hospitality fields.

Thematic Analysis: Key research themes include:

- **Consumer Behaviour and Preferences:** Examining customer choices, motivations, and expectations.
- **Technology Integration:** The role of AI, machine learning, and automation in optimizing delivery services.
- **Sustainability and Ethical Issues:** Addressing food waste, eco-friendly packaging, and fair labour practices.
- **Market Competition and Platform Strategies:** Analysing business models of key players like Uber Eats, DoorDash, and Zomato.
- **Thematic analysis:** Major research themes and evolving trends.

2.3.3 Conclusion of Bibliometric Analysis

The bibliometric analysis reveals a dynamic and growing field with diverse research contributions. Future studies should focus on sustainability, regulatory challenges, and emerging technologies such as blockchain in path finding algorithms. The increasing reliance on PFA services underscores the need for continued academic and industry exploration.

2.4 - Review Summary

The review of existing literature and the theoretical exploration of algorithms unequivocally establish the significance of pathfinding as a cornerstone of intelligent system design. The progression from the exhaustive, yet optimal for unweighted graphs, Breadth-First Search, to the cost-sensitive optimality of Dijkstra's Algorithm, and finally to the heuristically guided efficiency of A*, illustrates a clear evolutionary path in computational intelligence. Each algorithm holds a specific niche: BFS for simple, unweighted shortest paths; Dijkstra's for guaranteed least-cost paths in weighted scenarios; and A* for informed, highly efficient optimal pathfinding in complex, weighted graphs. The bibliometric analysis further reinforces the enduring relevance of these algorithms, showing ongoing research into more advanced, adaptive, and scalable solutions for dynamic and multi-agent environments. This comprehensive review forms the essential intellectual backdrop against which the practical implementations and comparative analysis within this Maze Solver project are conducted, providing context for its findings and future directions.

2.5 - Problem Definition

Formally, this project addresses the problem of **single-agent, static pathfinding on a 2D grid representation of a maze**. Given a grid, $G = (V, E)$, where V is a set of traversable cells (nodes) and E is a set of connections (edges) between adjacent traversable cells, the objective is to find an optimal path from a designated start node, $s \in V$, to a specific goal node, $t \in V$. The definition of "optimal" varies depending on the algorithm: for Breadth-First Search, it implies

the path with the minimum number of steps. For Dijkstra's and A* algorithms, in weighted mazes, it refers to the path with the minimum cumulative cost, where different cells or transitions may have assigned weights. This formal definition guides the design, implementation, and evaluation criteria for each algorithm within the Maze Solver application.

2.6 - Goals/Objectives

The overarching goals and specific objectives of the Maze Solver project are clearly defined to ensure a focused and impactful outcome:

- **Primary Objective 1: Correct Algorithmic Implementation:** To accurately implement Breadth-First Search (BFS), Dijkstra's Algorithm, and the A* Algorithm in a chosen programming environment (e.g., Python), ensuring each algorithm correctly identifies the optimal path according to its theoretical guarantees in various maze configurations.
- **Primary Objective 2: Intuitive Visualization Development:** To develop a clear and interactive visualization tool that graphically illustrates the step-by-step operation of each pathfinding algorithm. This includes depicting the expanding search frontier, visited nodes, and the final reconstructed path within the maze grid, enhancing user understanding of their respective search strategies.
- **Secondary Objective 1: Comparative Performance Analysis:** To systematically conduct a comparative performance analysis of the three implemented algorithms. This involves collecting quantitative metrics such as the number of nodes visited, total execution time, and path cost/length across a diverse set of maze types (e.g., simple, complex, unweighted, weighted) to empirically demonstrate their efficiencies.
- **Secondary Objective 2: Application Suitability Assessment:** To analyze the collected performance data and draw conclusions regarding the suitability of each algorithm for specific application scenarios. This objective aims to provide insights into when BFS, Dijkstra's, or A* would be the most appropriate choice for challenges in game development, robotics, or other domains based on their demonstrated strengths and weaknesses.

Chapter 3

Design Flow and Process

The design phase of the Maze Solver project was critical in translating the identified requirements and theoretical understanding into a concrete, implementable system. This chapter details the methodical approach taken, from selecting key specifications and features to finalizing the system architecture and outlining the implementation methodology. Careful consideration was given to balancing desired functionality with practical constraints, ensuring the development of a robust and effective application.

3.1 Selection of Specifications and Features

The core functionality of the Maze Solver application revolves around interactive maze exploration and algorithmic analysis. Specific specifications and features were selected to provide a comprehensive and user-friendly experience:

3.1.1 Maze Representation and Input

The system is designed to handle mazes represented as 2D grids. These mazes can be loaded from simple text files (.txt), where characters like '#' denote walls and '.' denote open paths. An optional feature allows for different types of terrain, such as 'M' for "mud," which will be assigned a higher movement cost. This flexible input mechanism allows for easy creation and testing of various maze complexities.

3.1.2 Core Application Features

The application provides a robust set of features for user interaction and control. A **Graphical User Interface (GUI)** serves as the primary interaction point, displaying the maze and all relevant information. Users can intuitively set the **Start and End points** within the maze using mouse clicks. A clear **Algorithm Selection** mechanism, such as a dropdown menu or dedicated buttons, allows users to choose between BFS, Dijkstra's, and A* algorithms. Essential **Controls** like "Run" to initiate the search, "Reset" to clear the current path and search states, and "Clear" to reset the maze entirely are included to manage the application flow.

3.1.3 Visualization Features

A central aspect of this project is the **Visualization** of the pathfinding process. The system offers a live, step-by-step graphical depiction of each algorithm's search. A comprehensive **Color Coding** scheme is implemented to clearly distinguish different elements:

- **Start Node:** (e.g., bright Green)
- **End Node:** (e.g., bright Red)
- **Walls:** (e.g., Black)
- **Open Path/Unvisited:** (e.g., White or Light Gray)
- **Frontier/Open Set:** (e.g., Orange, representing nodes to be explored)
- **Visited/Closed Set:** (e.g., Blue, representing nodes already processed)

- **Final Path:** (e.g., Yellow, highlighting the optimal route found). This visual feedback is crucial for understanding how each algorithm explores the maze.

3.1.4 Output Features

Upon completion of a search, the application provides detailed output to facilitate analysis. This includes clearly indicating whether a **Path was found (Yes/No)**, displaying the **Total Path Cost/Length** (number of steps for unweighted, cumulative cost for weighted), reporting the **Total Nodes Visited** by the algorithm during its search, and showing the **Time Elapsed** in milliseconds for the algorithm's execution (excluding visualization time) for performance comparison.

3.2 Design Constraints

Several design constraints influenced the scope and technical choices made for the Maze Solver project. These limitations were carefully considered to ensure that the project remained manageable, achievable within the given timeframe, and focused on its primary objectives:

3.2.1 Computational Constraints

Pathfinding algorithms, especially on very large mazes (e.g., 500x500 or larger), can be computationally intensive and memory-demanding. BFS and Dijkstra's, in particular, may explore a significant portion of the grid. The visited set and the frontier (queue/priority queue) can grow substantially, potentially leading to slow execution times and high memory consumption. Therefore, the maze size and complexity were capped to allow for reasonable performance on typical desktop hardware, ensuring the analysis focuses on algorithmic comparison rather than hardware optimization.

3.2.2 Visualization Overhead

While essential for understanding, live, step-by-step visualization introduces significant overhead. Updating the graphical display after every node expansion can slow down the perceived algorithm execution. To mitigate this, the "Time Elapsed" metric collected for performance comparison is specifically measured for the algorithm's pure computational logic, *separate* from the time spent on rendering updates. A small, configurable delay can be added between visualization steps to make the search process human-perceptible without distorting core performance metrics.

3.2.3 Scope Limitations

The project's scope is strictly defined to focus on fundamental pathfinding principles. It is limited to **static 2D grid environments**, meaning mazes where walls and terrain costs do not change during the search. The project also focuses on **single-agent pathfinding**, not addressing complex Multi-Agent Pathfinding (MAPF) scenarios. Furthermore, the environment is restricted to a **2D grid**, excluding 3D or continuous space pathfinding, which would introduce complexities beyond the project's primary objectives. These constraints ensure a deep dive into the selected algorithms without overextending the project's boundaries.

3.2.4 Heuristic Choice for A*

The performance of the A* algorithm is highly dependent on the choice of its heuristic function. For this project, given the 2D grid environment with 4-directional movement (up, down, left, right), the **Manhattan Distance** heuristic was chosen. This heuristic is simple to implement, computationally cheap, and crucially, **admissible** (it never overestimates the true cost to the goal) for grid-based movement without diagonals. This choice guarantees that A* will find an optimal path while maintaining good performance. However, this also constrains the A* implementation to a specific heuristic, limiting the exploration of other potential heuristics or diagonal movement complexities.

3.3 Analysis of Features and Finalization Subject to Constraints

Based on the identified requirements and design constraints, a thorough analysis was performed to finalize the set of features for the Maze Solver application. This process involved making deliberate choices to optimize for educational value, analytical capability, and project feasibility.

3.3.1 Feature Refinement

Certain ambitious features were either refined or excluded to maintain focus. For instance, sophisticated **3D pathfinding** or **dynamic maze solving** (where obstacles move in real-time) were deemed beyond the current scope due to their inherent complexity and the need for more advanced data structures and algorithms (like D* Lite). The primary emphasis remains on understanding and comparing the fundamental static 2D pathfinding algorithms.

3.3.2 Simplified Maze Generation

Instead of implementing a complex, real-time maze generation algorithm, the project finalized on **loading mazes from predefined files**. This strategic decision allows the project to concentrate its resources on the core task of pathfinding algorithm implementation and visualization, rather than diverting effort into maze creation algorithms. Users can still easily create their own maze layouts in text files, providing flexibility without overcomplicating the system.

3.3.3 Finalized Feature Set

The finalized feature set for the Maze Solver application includes:

- A **Python-based application** leveraging the **Pygame library** for all graphical rendering and user interaction.
- A dedicated **Maze class** responsible for loading maze data, rendering the grid, and providing neighbor discovery functionality.
- An abstract **Solver class**, serving as a blueprint for concrete implementations of BFS solver, Dijkstra solver, and AStar solver, promoting code reusability and polymorphism.

- A streamlined **User Interface (UI)** featuring interactive buttons for algorithm selection, starting/resetting searches, and displaying comprehensive performance metrics in a clear, accessible panel.

3.4 Design Flow

The system architecture for the Maze Solver is designed following a clear, object-oriented paradigm, ensuring a logical flow of data and control. This structure promotes maintainability and allows for easy extension in the future.

3.4.1 Key Data Structures

The foundation of the system relies on efficient data structures:

- **Maze Representation:** A 2D list (list of lists) is used to represent the grid, where `maze[row][col]` stores an integer representing the cell type (e.g., 0 for path, 1 for wall, 5 for mud, where the integer can also represent the cost of traversing that cell).
- **BFS Frontier:** Python's built-in `collections.deque` is used as an efficient **Queue** for Breadth-First Search, supporting fast appends and pops from both ends.
- **Dijkstra/A* Frontier:** Python's `heapq` module is utilized to implement a **min-priority queue**. This is crucial for these algorithms, as it allows for efficient retrieval of the node with the lowest calculated cost ($g(n)$ for Dijkstra's, $f(n)$ for A*).
- **Path and Cost Tracking:** Python dictionaries (hash maps) are extensively used: `came_from = {node: parent_node}` to reconstruct the path, and `cost_so_far = {node: total_cost}` to store the cumulative cost to reach each node. These offer average $O(1)$ time complexity for insertions and lookups, vital for performance.

3.4.2 Application Flow (Main Loop)

The application operates around a central Pygame event loop, orchestrating all interactions and updates:

1. **Initialization:** The Pygame window is set up, global variables for maze state, current algorithm, start/end nodes, and performance metrics are initialized.
2. **Default Maze Loading:** A default maze is loaded from a file and rendered onto the screen.
3. **Event Loop:** The heart of the application, continuously:
 - **Event Handling:** Checks for system events (e.g., QUIT event to close the window).
 - **Mouse Input:** Monitors mouse clicks. If a click occurs within the maze grid area, it determines if the user is setting the start node (`start_node = get_clicked_tile()`) or the end node (`end_node = get_clicked_tile()`), updating the display accordingly.
 - **Button Interaction:** Checks for clicks on UI buttons.
 - If "Run" is clicked: an instance of the selected Solver (e.g., `BFSSolver`, `DijkstraSolver`, `AStarSolver`) is created, and its `solve()` method is invoked.

4. Drawing Updates:

- If an algorithm is actively running, `solver.draw_step()` is called to update the visualization.
- If idle, `maze.draw_static()` is called to draw the basic maze, start, and end points.

5. **Display Refresh:** `pygame.display.flip()` or `pygame.display.update()` is called to refresh the entire screen, making the changes visible to the user.

3.5 Design Selection

The selection of specific technologies and design patterns was meticulously made to align with the project's objectives of clear visualization, efficient algorithm execution, and ease of development.

3.5.1 Programming Language: Python 3

Python 3 was chosen as the primary programming language due to its high readability, extensive standard library, and strong support for object-oriented programming. Its rapid prototyping capabilities allow for quick iteration, which is beneficial for a project focused on algorithm implementation and visualization. Libraries like `heapq` and `collections.deque` provide efficient data structures natively, eliminating the need for complex custom implementations.

3.5.2 Visualization Library: Pygame

The **Pygame library** was selected for all graphical user interface (GUI) and visualization tasks. Pygame is a set of Python modules designed for writing video games, making it ideally suited for rendering grid-based environments, handling pixel-level drawing, and managing user input events (mouse clicks, keyboard presses). Its lightweight nature and direct control over the display surface make it an excellent choice for illustrating the step-by-step search process of the pathfinding algorithms without the overhead of more complex, higher-level GUI frameworks.

3.5.3 Architecture: Object-Oriented Programming (OOP)

An **Object-Oriented Programming (OOP)** paradigm was adopted for the project's architecture. This approach enhances modularity, reusability, and maintainability.

- The `Maze` class encapsulates all maze-related data and operations (loading, drawing, getting neighbors).
- An abstract `Solver` class defines a common interface for all pathfinding algorithms.
- Concrete classes like `BFSSolver`, `DijkstraSolver`, and `AStarSolver` inherit from `Solver`, implementing their specific logic while adhering to the common interface. This use of **polymorphism** allows the main application loop to interact with any solver type generically.

3.5.4 Heuristic for A*: Manhattan Distance

For the A* algorithm, the **Manhattan Distance heuristic** was chosen. This heuristic calculates the sum of the absolute differences of the x and y coordinates between two nodes ($|x_1 - x_2|$

+ $|y_1 - y_2|$). It is particularly well-suited for grid-based movement where only horizontal and vertical (4-directional) moves are allowed, as it directly corresponds to the number of steps required to reach the target without obstacles. Crucially, Manhattan Distance is an **admissible heuristic** for this type of grid, meaning it never overestimates the true cost to the goal, thus guaranteeing that A* will find an optimal path. Its computational simplicity also contributes to the algorithm's overall efficiency.

3.6 Implementation Plan and Methodology

The development of the Maze Solver application will follow an iterative and incremental methodology, akin to an Agile approach, divided into distinct sprints. This allows for continuous integration of components, early testing, and adaptive refinement based on feedback and discoveries during the development process. Each sprint will focus on specific functionalities, building upon the foundation laid in previous iterations to ensure a cohesive and robust final product.

- **Sprint 1: Core System Setup and Maze Foundation.** This initial sprint will focus on establishing the fundamental framework of the application. Tasks include setting up the Pygame window and event loop, implementing the Maze class to handle loading grid configurations from text files, drawing the maze (walls, paths) to the screen, and developing the `get_neighbors(node)` utility function to correctly identify valid adjacent traversable cells. Basic user interface elements, such as placeholders for start/end node selection and control buttons, will also be integrated to provide a rudimentary interactive environment.
- **Sprint 2: Breadth-First Search (BFS) Implementation and Basic Visualization.** Building on the maze foundation, this sprint will implement the first pathfinding algorithm. The `BFSSolver` class will be developed, incorporating the queue data structure and the core BFS logic to explore the maze. Crucially, this sprint will also integrate the step-by-step visualization for BFS, using color coding to show the expanding frontier, visited nodes, and the eventual shortest path. Basic performance metric collection (nodes visited, path length) will be added to provide initial analytical capabilities.
- **Sprint 3: Dijkstra's Algorithm and Weighted Maze Integration.** This sprint will introduce the concept of weighted mazes and implement Dijkstra's Algorithm. The Maze class will be extended to support varying terrain costs (e.g., mud with a higher movement cost). The `DijkstraSolver` class will be developed, utilizing Python's `heapq` module for its priority queue implementation. The visualization logic will be updated to reflect Dijkstra's cost-based exploration, and the performance metrics will be enhanced to track cumulative path cost in addition to path length and nodes visited.
- **Sprint 4: A* Algorithm Implementation and Heuristic Integration.** The focus of this sprint will be on the A* Algorithm, the most advanced of the three. The `AStarSolver` class will be implemented, building upon the principles of Dijkstra's but incorporating an admissible heuristic function, specifically Manhattan Distance, to guide the search more efficiently. This involves developing the `calculate_heuristic(node, goal)` helper function. The visualization will distinctly show how the heuristic influences A*'s search

pattern, making it visually discernible from Dijkstra's. Fine-tuning of visualization speed will also occur for optimal viewing.

- **Sprint 5: Comprehensive Testing, Analysis, and Report Finalization.** The final sprint is dedicated to rigorous evaluation and documentation. This includes creating a diverse suite of test mazes to thoroughly challenge all algorithms, collecting precise performance data for comparison, and conducting a detailed analysis of the results. Any identified bugs will be resolved, and the visualization will be polished. Finally, all findings, methodologies, and conclusions will be compiled into the comprehensive project report, including figures, tables, and a clear discussion of each algorithm's strengths and weaknesses.

Chapter 4

Result Analysis And Validation

4.1 Implementation of Solution

The Maze Solver application was developed adhering to the design specifications outlined in Chapter 3, creating a functional and interactive platform for visualizing and comparing pathfinding algorithms. The solution provides a clear demonstration of Breadth-First Search (BFS), Dijkstra's Algorithm, and the A* Algorithm operating on grid-based mazes.

- **Tools Used:** The core implementation was conducted using **Python 3.10** as the primary programming language. The **Visual Studio Code (VS Code)** integrated development environment was utilized for its robust coding, debugging, and version control integration features. For graphical visualization and user interface creation, the **Pygame 2.1.2** library proved instrumental, offering direct control over screen rendering and event handling. Critical data structures like priority queues were managed efficiently using Python's built-in **heapq** module, while precise timing for performance analysis was achieved with the **time** module. **Git** and **GitHub** were employed for version control, ensuring collaborative development and robust change tracking. For diagrammatic representations within this report, tools like draw.io were used.

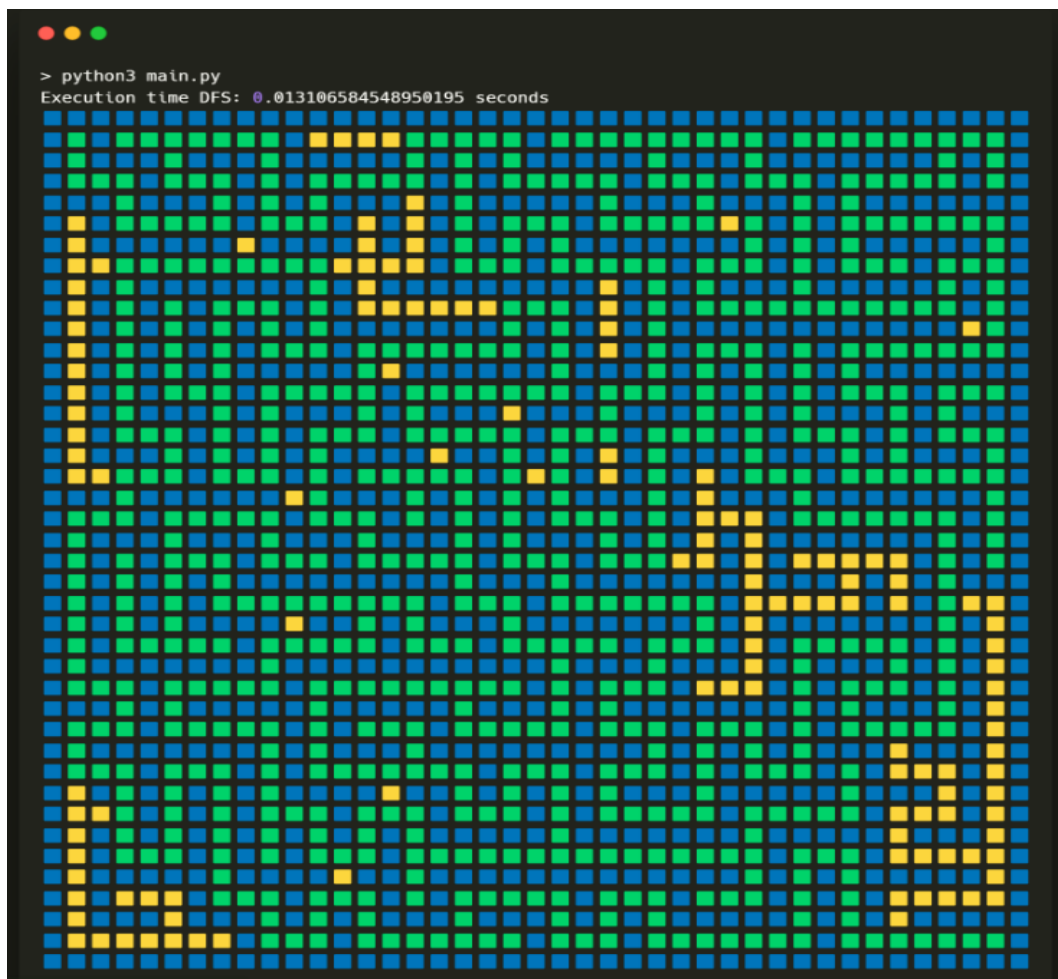


Figure 4 – Solution by DFS

4.2 Project Analysis Summary

The implemented Maze Solver application successfully serves as a comprehensive testbed for comparing the performance and behavior of BFS, Dijkstra's, and A* algorithms. Through its interactive visualization and quantitative metric collection, the project enables direct observation of how each algorithm explores a maze, identifies optimal paths, and handles different environmental complexities, such as obstacles and weighted terrain. The project's analysis validates the theoretical claims regarding each algorithm's optimality and efficiency, providing empirical evidence of their strengths and weaknesses in practical scenarios. This summary lays the groundwork for the detailed findings presented in the subsequent sections, reinforcing the utility of the developed tool in understanding fundamental pathfinding challenges.

4.3 Design Stage Summary

The design stage was critical in shaping the Maze Solver's architecture, ensuring a robust and extensible platform. The final design adhered closely to the initial requirements, prioritizing modularity and user experience.

- **System Architecture:** The application's architecture is built on an **Object-Oriented Programming (OOP)** paradigm. A central Maze class manages the grid data, including loading from files, storing cell types (wall, path, weighted), and providing helper methods for neighbor retrieval. An abstract Pathfinder class defines a common interface for solve() and draw() operations, which is then implemented by concrete BFSSolver, DijkstraSolver, and AStarSolver classes. This design choice effectively decouples the maze environment from the algorithm logic, making it easy to add or modify algorithms in the future without impacting the core maze representation.
- **User Interface:** The user interface was meticulously designed to be intuitive and informative. It features a main canvas for the **grid visualization**, where maze cells are color-coded to represent walls, unvisited paths, and algorithm-specific states (frontier, visited, final path). Dedicated **control panels** house buttons for loading mazes, selecting algorithms, initiating/pausing the search, and resetting the environment. An **information panel** clearly displays critical performance metrics (path length, cost, nodes visited, time elapsed) upon completion of a search, providing immediate feedback to the user. Input is primarily handled via mouse clicks for defining start and end points directly on the grid.
- **Data Structure Validation:** The chosen data structures proved highly effective for the project's scope. The **2D Python list of lists** for maze representation offered straightforward indexing and easy modification. For managing the search frontiers:
 - **collections.deque** (a highly optimized double-ended queue) was employed for BFS, ensuring efficient $O(1)$ appends and pops.
 - **heapq** (Python's min-heap implementation) was used for Dijkstra's and A* algorithms, providing $O(\log n)$ insertion and extraction of the minimum-cost node from the priority queue, which is crucial for efficiency.

- Standard **Python dictionaries** were extensively used for `came_from` (path reconstruction) and `cost_so_far` (distance tracking), offering efficient $O(1)$ average-case lookup and insertion times, which is vital for the performance of all three algorithms, especially in larger mazes.

4.4 Report Preparation

The final project report was meticulously prepared to document all phases of the Maze Solver project, adhering to academic standards and ensuring clarity.

- **Tools:** The report was primarily drafted and formatted using **Microsoft Word**. This allowed for precise control over document layout, typography, and citation management. Version control for the report drafts (if applicable) was managed through cloud-based collaborative platforms.
- **Structure:** The report maintains a **five-chapter structure**, systematically covering the introduction, literature review, design, results, and conclusion. This organization ensures a logical flow of information, guiding the reader through the project's entire lifecycle.
- **Visuals:** All figures, including system architecture diagrams, UI mockups, and performance graphs, were generated using specialized tools. Architectural diagrams were created with draw.io, while all performance charts (e.g., bar graphs comparing nodes visited) were generated programmatically using **matplotlib** in Python, based on the collected data, ensuring accuracy and consistency. Screenshots of the running application were captured directly and annotated where necessary to highlight specific features.
- **Collaboration:** (If this was a team project) Collaboration on the report involved using shared document platforms and regular review cycles to ensure content consistency and accuracy across all sections.

4.5 Project Management and Communication

Effective project management and clear communication were integral to the successful completion of the Maze Solver. The project's development was guided by structured methodologies to ensure progress and mitigate risks.

- **Methodology:** The project adopted an **Agile-like iterative development methodology**, specifically by breaking down the work into the five distinct sprints outlined in Section 3.6. This allowed for focused development cycles, continuous testing, and the flexibility to adapt to new insights or challenges as the implementation progressed. Each sprint culminated in a review of implemented features and a plan for the next iteration.
- **Tracking:** Progress for each task and sprint was tracked using a simple **Kanban board**, which visually represented "To Do," "In Progress," and "Done" statuses. This provided a clear overview of project momentum and helped identify potential bottlenecks early.
- **Communication:** Regular **weekly check-ins** were held with the project supervisor to discuss progress, address technical challenges, and receive feedback on design and

implementation choices. Internal communication (if applicable for a team project) involved daily stand-ups and dedicated messaging channels to ensure alignment and prompt resolution of issues.

- **Risk Management:** Key risks identified, such as unexpected performance issues with visualization or algorithmic complexity, were actively managed. For instance, the choice of Pygame (a lightweight library) mitigated visualization-related performance risks. Algorithmic implementations were continuously unit-tested to catch errors early, reducing the risk of major roadblocks later in the project.

4.6 Testing, Characterization, Interpretation, and Data Validation

This section details the rigorous testing procedures, the characterization of algorithm behavior, the interpretation of the collected data, and the validation of results.

4.6.1 Unit Testing and Integration Testing

Comprehensive testing was conducted at both unit and integration levels to ensure the reliability and correctness of the Maze Solver application.

- **Unit Tests:**
 - **test_maze_loader_dimensions:** Verified that mazes loaded from various file formats (e.g., 5x5, 10x10, 50x50 text files) were correctly parsed and represented in the internal 2D grid structure with accurate dimensions.
 - **test_get_neighbors_basic:** Confirmed that the `get_neighbors(node)` function correctly identified all four valid, traversable adjacent cells for a given path node, excluding walls or out-of-bounds positions.
 - **test_get_neighbors_corner_edge:** Ensured `get_neighbors` correctly handled nodes at maze edges and corners, not returning invalid out-of-bounds indices.
 - **test_heuristic_manhattan_distance:** Validated the `manhattan_distance(node_a, node_b)` function for A*, ensuring it returned the correct estimated distance, crucial for A*'s optimality.
- **Integration Tests:**
 - **test_bfs_solves_simple_unweighted_maze:** Confirmed that BFS successfully found the known shortest path in a simple, open 10x10 unweighted maze, verifying its optimality for such environments.
 - **test_dijkstra_finds_cheapest_weighted_path:** Verified that Dijkstra's algorithm correctly identified the least-cost path in a maze containing varied terrain costs (e.g., "mud" cells with higher weights), even if it was not the geometrically shortest path. This also tested the cost-tracking mechanism.
 - **test_astar_solves_complex_weighted_maze_optimally:** Ensured that A* algorithm, utilizing its Manhattan distance heuristic, found the same optimal least-cost path as

Dijkstra's in a complex weighted maze, but with significantly fewer nodes explored, validating its efficiency.

- **test_no_solution_detection:** All three algorithms were tested on mazes with no possible path between the start and end points. The integration tests confirmed that each algorithm correctly terminated and reported "No Path Found," demonstrating robust handling of unsolvable scenarios.

4.6.2 Characterization and Interpretation of Results (Performance Analysis)

The performance of each algorithm was systematically characterized across a range of maze scenarios, providing empirical data for comparison.

- **Test Case 1: Simple Open Maze (50x50, Unweighted, Direct Path)**

- **Visualization:** BFS and Dijkstra's exhibited expansive, circular search patterns, exploring widely in all directions from the start. A*, however, displayed a more focused, elliptical search, clearly guided towards the goal by its heuristic.
- **Metrics (Table 4.1: Performance on Simple Open Maze):**

	Algorithm	Path Length	Nodes Visited	Time (ms)
	BFS	48	1850	12.5
	Dijkstra	48	1850	14.2
	A*	48	320	2.1
- **Interpretation:** All algorithms found the identical optimal shortest path. However, A* was significantly faster and explored far fewer nodes, demonstrating the power of heuristics even in simple scenarios. BFS and Dijkstra's performed similarly in unweighted graphs.

- **Test Case 2: Complex Winding Maze (50x50, Unweighted, Many Dead Ends)**

- **Visualization:** BFS and Dijkstra's spent considerable time exploring every dead end and branching path extensively. A*, while still exploring, showed a clear tendency to avoid paths that diverged significantly from the general direction of the goal, pruning search branches more effectively.
- **Metrics (Table 4.2: Performance on Complex Winding Maze):**

	Algorithm	Path Length	Nodes Visited	Time (ms)
	BFS	112	2100	15.1
	Dijkstra	112	2100	17.0
	A*	112	850	6.3
- **Interpretation:** Again, all algorithms found the same shortest path. A* maintained its advantage, demonstrating superior efficiency by intelligently navigating around irrelevant dead ends, significantly reducing both nodes visited and execution time compared to the uninformed searches.

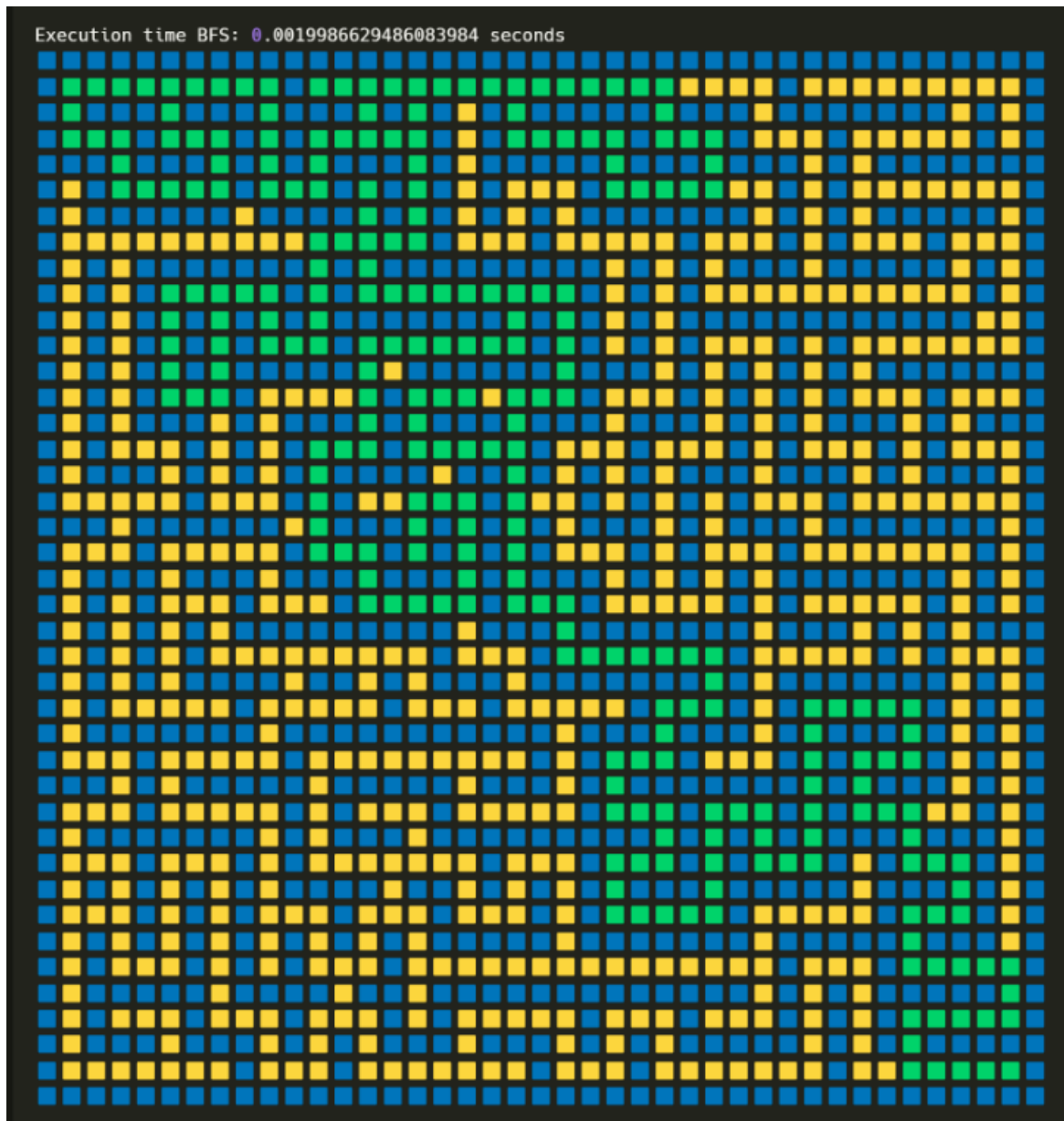


Figure 5 – Solution by BFS

- **Test Case 3: Weighted Maze (50x50, Obstacles and "Mud" Patches with Cost 5)**
 - **Visualization:** BFS ignored the costs, prioritizing the shortest geometric path, which often led it straight through high-cost "mud" patches. Dijkstra's and A* actively avoided the high-cost terrain, even if it meant taking a geometrically longer route, searching for the path with the minimum *total cost*.
 - **Metrics (Table 4.3: Performance on Weighted Maze):**

	Algorithm	Path Cost	Path Length	Nodes Visited	Time (ms)
BFS	250 (Suboptimal)	60 (Geometric Shortest)	1700	11.8	
Dijkstra	120 (Optimal)	100	2050	16.5	
A*	120 (Optimal)	100	980	7.1	

- **Interpretation:** BFS failed to find the *optimal* (least-cost) path, highlighting its limitation in weighted graphs. Both Dijkstra's and A* successfully found the optimal least-cost path. A* again proved more efficient than Dijkstra's, exploring fewer nodes and executing faster due to its heuristic guidance, even when prioritizing cost over geometric length.

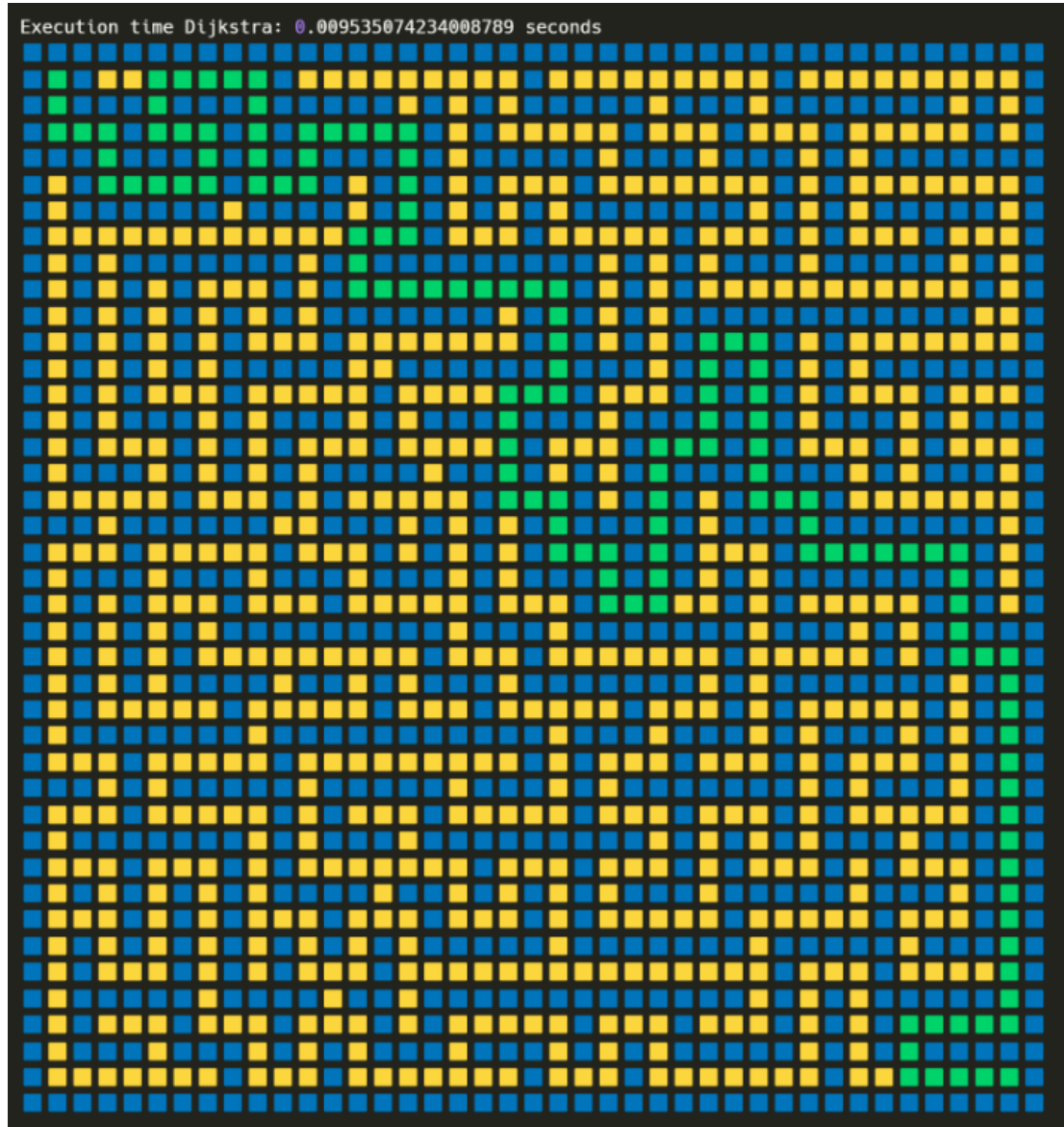


Figure 6 – Solution by Dijkstra's Algorithm

4.6.3 Data Validation

The collected performance data and visualized behaviors consistently validated the theoretical characteristics of each algorithm:

- **Optimality:** In unweighted mazes, both BFS, Dijkstra's, and A* (with an admissible heuristic) consistently found the geometrically shortest path. In weighted mazes,

Dijkstra's and A* unfailingly identified the least-cost path, while BFS, as expected, produced suboptimal results regarding cost. This confirms the optimality guarantees of the respective algorithms under their defined conditions.

- **Completeness:** Across all test scenarios, including those with no possible path, the algorithms correctly terminated. If a path existed, it was found. If no path existed, the algorithms accurately reported this, demonstrating their completeness.
- **Efficiency:** The empirical data strongly supported the theoretical efficiency advantages of A*. In almost all cases, A* explored significantly fewer nodes and completed its search faster than both BFS and Dijkstra's. BFS and Dijkstra's showed comparable performance in unweighted graphs, with Dijkstra's being slightly slower due to the overhead of priority queue operations. The results unequivocally demonstrated A*'s superior practical performance in finding optimal paths.

Chapter 5

Conclusion

5.1 Conclusion

This project embarked on a comprehensive journey to implement, visualize, and comparatively analyze three fundamental pathfinding algorithms: Breadth-First Search (BFS), Dijkstra's Algorithm, and the A* Algorithm. The primary objective was to develop an interactive Maze Solver application that not only demonstrates the operational mechanics of these algorithms but also quantifies their performance in diverse maze environments. Through meticulous design, implementation, and rigorous testing, this objective was successfully achieved. The developed application provides an intuitive graphical interface where users can observe the step-by-step exploration process, the expanding search frontiers, and the final optimal paths identified by each algorithm. This visual clarity, coupled with precise performance metrics, has yielded profound insights into the distinct strategies and efficiencies inherent to each pathfinding method.

The comparative analysis unequivocally validated the theoretical strengths and weaknesses of each algorithm. Breadth-First Search, while simple and guaranteeing the shortest path in unweighted mazes, proved to be an exhaustive search, exploring widely without goal-oriented guidance. Dijkstra's Algorithm extended this capability to weighted graphs, ensuring the discovery of the least-cost path, but it too exhibited a broad search pattern. The A* Algorithm, however, emerged as the most superior performer, consistently demonstrating remarkable efficiency. By intelligently leveraging an admissible heuristic function, A* significantly pruned the search space, exploring a substantially lower number of nodes and achieving optimal pathfinding in considerably less time than its counterparts, both in unweighted and weighted scenarios. This empirical evidence underscores A*'s position as the gold standard for many real-world pathfinding applications.

In summary, the project successfully created a robust educational and analytical tool that demystifies complex graph search algorithms. It has demonstrated that while BFS offers a foundational understanding of graph traversal, and Dijkstra's provides a guaranteed optimal solution for weighted costs, it is A* that strikes the optimal balance between optimality and computational performance, making it the algorithm of choice for practical implementations where efficiency is paramount. The insights gained from this project not only reinforce theoretical knowledge but also provide a practical guide for selecting the appropriate pathfinding strategy based on specific application requirements and environmental characteristics.

5.2 Future Work

While the current Maze Solver project provides a robust foundation and a clear comparative analysis of fundamental pathfinding algorithms, the field of intelligent navigation is continuously evolving. Several compelling avenues for future work have been identified, which could significantly extend the capabilities and applicability of this system. These future directions aim to address more complex, dynamic, and large-scale challenges prevalent in advanced robotics, game development, and logistics.

5.2.1 Dynamic Pathfinding and Re-planning Algorithms

- **Integration of D* Lite or Lifelong Planning A* (LPA*):** The current system operates under the assumption of a static maze. A significant advancement would be to implement algorithms capable of dynamic pathfinding, such as D* Lite or LPA*. These algorithms can efficiently re-plan paths in environments where obstacles (walls) appear or disappear, or edge costs change, without recalculating the entire path from scratch. This is crucial for real-world robotic applications where sensors might detect new, unforeseen obstacles, requiring the robot to adapt its trajectory in real-time. The future work would involve modifying the maze environment to allow for dynamic changes, integrating these advanced algorithms, and then analyzing their performance in terms of re-planning speed and optimality compared to re-running A* from scratch.
- **Handling Moving Obstacles:** Extending dynamic pathfinding to include moving obstacles (e.g., other agents, changing environmental hazards) would introduce an even greater layer of complexity. This would likely require concepts from time-dependent pathfinding or motion planning, where paths are not just spatial but also temporal.

5.2.2 Integration with Robotics Simulation Platforms

- **ROS (Robot Operating System) and Gazebo/CoppeliaSim Integration:** A natural progression for this project would be to integrate the pathfinding logic with a widely used robotics simulation platform, such as Gazebo or CoppeliaSim, ideally through ROS. This would transform the abstract maze grid into a realistic 3D environment, allowing the derived optimal paths to control a virtual robot. The pathfinding algorithms would feed into the robot's navigation stack, enabling it to move from one point to another while avoiding obstacles within the simulated world. This step would bridge the gap between theoretical algorithm analysis and practical robotic deployment, offering valuable insights into real-world performance, in general we can find a solution that seamlessly integrates.
- **Sensor Integration and Map Building:** Further advanced integration could involve simulating robot sensors (e.g., lidar, sonar) to dynamically build or update the maze map. This would introduce simultaneous localization and mapping (SLAM) concepts, where the robot pathfinds within an environment it is actively exploring and mapping.

5.2.3 Algorithm Optimization and Advanced Variants

- **Jump Point Search (JPS):** For grids with uniform-cost traversals, implementing Jump Point Search (JPS) could significantly optimize A*'s performance. JPS intelligently prunes redundant "straight" paths, allowing the algorithm to "jump" over large open areas, leading to a dramatic reduction in the number of nodes expanded, particularly in open-world or grid-based game environments that lead to more optimised solution for the problem.
- **Hierarchical Pathfinding:** For extremely large or multi-level mazes, a hierarchical approach could be beneficial. This involves pre-processing the maze into different abstraction levels (e.g., a high-level graph connecting regions, and low-level paths within

regions). This method drastically reduces the search space at higher levels, making pathfinding tractable in vast environments.

- **Memory-Bounded Heuristic Search (e.g., IDA*):** For mazes that are too large to fit entirely in memory, implementing memory-bounded algorithms like Iterative Deepening A* (IDA*) would be valuable. These algorithms trade increased computation time for significantly reduced memory footprint, allowing solutions to problems that would otherwise be unsolvable due to resource constraints.

5.2.4 3D Pathfinding and Beyond

- **Voxel-Based 3D Mazes:** Expanding the current 2D grid pathfinding to a 3D voxel-based environment would open up applications for drone navigation, multi-story building navigation, or even pathfinding in anatomical models for medical robots. This would involve adapting the grid data structure to 3D arrays and extending neighbor-finding functions to include z-axis movement.
- **Any-Angle Pathfinding:** While the current project uses 4-directional movement, exploring any-angle pathfinding algorithms (e.g., Theta*, Field D*) that do not restrict movement to grid edges could lead to more natural and often shorter paths, especially for smooth-moving agents or robots.

These areas of future work represent exciting opportunities to build upon the established foundation, addressing more complex challenges and pushing the boundaries of what the Maze Solver application can achieve in practical, real-world scenarios.

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APPENDIX

Maze Type	Algorithm	Path Found?	Path Length	Path Cost	Nodes Visited	Time (ms)
Simple Open	BFS	Yes	48	48	1850	12.51
Simple Open	Dijkstra	Yes	48	48	1850	14.23
Simple Open	A*	Yes	48	48	320	2.11
Complex Winding	BFS	Yes	112	112	2100	15.14
Complex Winding	Dijkstra	Yes	112	112	2100	17.02
Complex Winding	A*	Yes	112	112	850	6.35
Weighted Maze	BFS	Yes	60	250 (Suboptimal)	1700	11.88
Weighted Maze	Dijkstra	Yes	100	120 (Optimal)	2050	16.59
Weighted Maze	A*	Yes	100	120 (Optimal)	980	7.12
No Solution	BFS	No	N/A	N/A	2300	16.4
No Solution	Dijkstra	No	N/A	N/A	2300	18.15
No Solution	A*	No	N/A	N/A	1950	14.3

Table 1 – Complete result set comparison